

Omneya Osama

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EDUCATION

Misr International University (MIU)

Bachelor of Science in Computer Science, Minor in Artificial Intelligence

Cairo, Egypt

2022 – 2026

EXPERIENCE

National Bank of Kuwait HQ

HR Intern

Aug. 2025

Cairo, Egypt

- Supported recruitment operations and employee engagement initiatives within the HR department.
- Assisted in organizing and maintaining candidate records while ensuring compliance with HR policies.
- Coordinated with HR teams to streamline onboarding workflows and improve communication processes.

Envnt

Software Development Intern

Jul. 2025

Cairo, Egypt

- Built and deployed two applications using OutSystems: an Employee Directory and a Tech E-Commerce website.
- Applied low-code development practices to design responsive frontends and implement database-driven features.
- Collaborated with the development team to optimize application workflows, improving usability and scalability.

PROJECTS

ReVerie – AI-Powered VR Memory Reconstruction Platform

Sep. 2025 – Present

Python, Computer Vision, Deep Learning, GCP, Supabase, Unreal Engine, Flutter

- Co-developing an AI-powered VR therapeutic platform within a multidisciplinary team to transform personal photographs into immersive reminiscence experiences for Alzheimer's patients.
- Contributed to memory reconstruction workflows using depth estimation, segmentation, inpainting, and 3D reconstruction of humans, objects, and environments.
- Developed scalable GCP-based AI pipelines integrated with Supabase and contributed to Flutter mobile interfaces and Unreal Engine visualization for supervised patient-doctor interactions.

Smart Bin Waste Management System

May 2026 - Jun. 2026

TensorFlow/Keras, DenseNet121, OpenCV, Arduino, A Search*

- Built a multi-agent waste management prototype combining waste classification, fill-level monitoring, overflow prediction, and route optimization.
- Fine-tuned DenseNet121 on a 2,527-image, 4-class garbage dataset, achieving 98% test accuracy and 97.89% macro F1-score.
- Integrated A* search algorithm for intelligent waste pickup route planning.

Multi-Label Emotion Intelligence System

May 2025

PyTorch, Hugging Face, RoBERTa, BERT, NLP

- Fine-tuned transformer-based models for multi-label emotion classification using class-weighted BCE loss and per-label threshold optimization.
- Achieved 0.598 Micro-F1, 0.964 Hamming Accuracy, and 0.555 Sample Jaccard on the GoEmotions dataset.
- Developed a dedicated sarcasm detection model achieving an F1-score of 0.635.

Real-Time Sign Language Recognition

Apr. 2025 – May 2025

Python, YOLOv8, OpenCV, Roboflow

- Built a real-time ASL recognition system using YOLOv8 and OpenCV for live webcam-based inference.
- Trained a YOLOv8 model on a custom ASL dataset with A–Z hand sign annotations.
- Developed pipelines for static hand sign recognition and real-time letter spelling.

TECHNICAL SKILLS

Languages: C/C++, Java, Python, SQL, Assembly, JavaScript, HTML/CSS

Web/Backend: PHP, Node.js, Express.js, MongoDB, MySQL, XAMPP

AI/ML: TensorFlow, OpenCV, MediaPipe, YOLO, BERT, Scikit-learn, PyTorch, Hugging Face

Tools/Platforms: Git, GitHub, VS Code, Visual Studio, Google Colab, Jupyter Notebook, OutSystems

Systems: Linux, Ubuntu, Virtual Machines

EXTRACURRICULAR ACTIVITIES

Co-Assistant – Data Structures and Algorithms Course

Assisted students with core algorithms, data structures, assignments, and coding support.